The Fig. 4 Potential Distribution of Nakagawa 1990, Solved Four Ways

Finite Volumes, Conformal Mapping and a Green's Function for a Thin-SOI High-Voltage Device

Structure B, 500 V — theory, algorithm, formulae and plots

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Abstract

Figure 4 of Nakagawa *et al.* (ISPSD 1990) shows that when a thin (20 µm) silicon film on a buried oxide (BOX) fully depletes, more than half of the applied 500 V is forced across the thin oxide. This report reproduces that two-dimensional potential map by *four independent methods* and cross-validates them: (i) a 1-D vertical model, (ii) a finite-volume Poisson solve, (iii) an analytic *conformal map* (Schwarz–Christoffel, Jacobi elliptic functions), and (iv) an analytic *Green's function* (rectangle eigenfunction series). For each analytic method we give the full derivation, the governing formulae, the algorithm, and the generated plot. The two analytic methods agree to 0.09 V RMS (0.02% of V) — both predict an oxide voltage share of $\approx 58\%$ — and both track the finite-volume solve (54%) to within the boundary-condition idealisation they share. The charge-free limit is shown to give only $\approx 31\%$, isolating the donor space charge as the origin of the "more than half" result.

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1 Device, physics and governing equations

1.1 The structure

Structure B is a lateral cross-section (Fig. 1): a silicon drift film of thickness $t_{\mathrm{si}} = 20\,\mu\mathrm{m}$ sits on a buried oxide of thickness $t_{\mathrm{ox}} = 3\,\mu\mathrm{m}$ over a grounded substrate. A drain electrode ($\phi = V$) and

a source electrode ($\phi = 0$) occupy the top surface; a p^+ sinker grounds the right silicon edge. A shallow N donor sheet (dose $S_D = 1.5 \times 10^{12}\,\mathrm{cm}^{-2}$ over the bottom $w_n = 2\,\mu\mathrm{m}$ of the film, atop a background $N_{D0} = 1 \times 10^{14}\,\mathrm{cm}^{-3}$) depletes and shields the bulk silicon. All lengths are quoted in µm; the code works in CGS-practical units (cm, V, cm^{-3}).

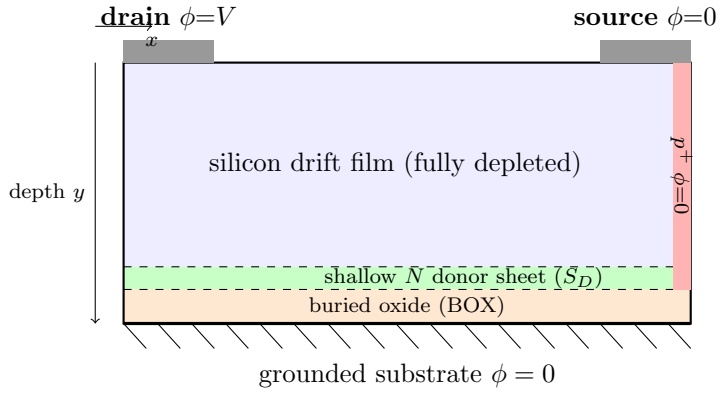


Figure 1: Structure B cross-section (not to scale). The fully-depleted silicon film sits on the buried oxide over a grounded substrate; the shallow donor sheet shields the bulk silicon, and the p^+ sinker grounds the right edge.

Table 1: Model parameters (identical across all four methods).

t_{si}	20 µm	drain / source width	8 µm
t_{ox}	3 µm	w_n (donor sheet)	2 µm
L_x	50 µm	N_{D0}	$1 \times 10^{14}\,\mathrm{cm}^{-3}$
V	500 V	S_D	$1.5 \times 10^{12}\,\mathrm{cm}^{-2}$
$\varepsilon_{\mathrm{si}} = 11.7\,\varepsilon_0$	$\varepsilon_{\mathrm{ox}} = 3.9\,\varepsilon_0$	$q = 1.602 \times 10^{-19}$ C	$\varepsilon_0 = 8.854 \times 10^{-14}\,\mathrm{F\,cm}^{-1}$

1.2 Poisson's equation with two dielectrics

With x lateral and y the depth (downward), the electrostatic potential $\phi(x, y)$ obeys

$$\nabla\cdot\big(\varepsilon(x,y)\,\nabla\phi\big) = -\rho(x,y), \qquad \rho = q\,N(y) \text{ in Si}, \quad \rho = 0 \text{ in oxide}, \tag{1}$$

with $\varepsilon = \varepsilon_{\mathrm{si}}$ in the silicon ($0 \le y < t_{\mathrm{si}}$) and $\varepsilon = \varepsilon_{\mathrm{ox}}$ in the oxide ($t_{\mathrm{si}} \le y < H$), $H = t_{\mathrm{si}} + t_{\mathrm{ox}}$. Across the Si/oxide interface the potential and the normal displacement are continuous,

$$\phi|_{t_{\mathrm{si}}^-} = \phi|_{t_{\mathrm{si}}^+}, \qquad \varepsilon_{\mathrm{si}}\,\frac{\partial\phi}{\partial y}\Big|_{t_{\mathrm{si}}^-} = \varepsilon_{\mathrm{ox}}\,\frac{\partial\phi}{\partial y}\Big|_{t_{\mathrm{si}}^+}, \tag{2}$$

so the vertical field jumps by $\varepsilon_{\mathrm{si}}/\varepsilon_{\mathrm{ox}} = 3$ on entering the oxide — the physical origin of the oxide "voltage cliff." The boundary conditions for structure B are Dirichlet on the drain ($\phi = V$), source ($\phi = 0$), grounded substrate ($\phi = 0$) and p^+ edge ($\phi = 0$), and natural (zero-flux) Neumann on the remaining free surfaces.

1.3 The metric: oxide voltage share

The single number that encodes Fig. 4 is the fraction of V dropped across the oxide, evaluated on the vertical cut under the drain centre,

$$s_{\mathrm{ox}} \equiv \frac{\phi(x_d,\, t_{\mathrm{si}})}{\phi(x_d,\, 0)}, \qquad x_d = \tfrac{1}{2}\,(\text{drain width}), \tag{3}$$

because the potential still remaining at the interface must fall entirely across the grounded oxide. The paper states $s_{\mathrm{ox}} > \frac{1}{2}$.

2 Method 1 — finite-volume Poisson solve

The reference numerical solution discretises (1) directly. On a rectangular grid, a finite-volume (conservative) integration of each cell gives the five-point stencil

$$\sum_{f\in\{W,E,N,S\}} \frac{\kappa_f}{h_f^2}\,(\phi_f-\phi_P) = -\rho_P, \qquad \kappa_f = \frac{2\,\varepsilon_P\varepsilon_f}{\varepsilon_P+\varepsilon_f}, \tag{4}$$

where the *harmonic-mean* face permittivity κ_f is the exact one-dimensional series-capacitor value and reproduces the displacement jump (2) to second order when the interface lies on a cell face. Dirichlet nodes become identity rows. Assembled as a sparse matrix $\mathbf{A}\boldsymbol{\phi} = \mathbf{b}$ and solved by a sparse direct factorisation, this is `fig4_pde_standalone.py`. Richardson extrapolation of the drain-averaged share (3) over grids up to 251×231 gives a grid-converged $s_{\mathrm{ox}} \approx 0.54$ at second order. This method makes no approximation beyond discretisation and is the yardstick for the analytic methods.

3 Method 2 — conformal mapping

Conformal mapping solves Laplace's equation, but (1) is Poisson with two dielectrics. Two standard, clearly-scoped reductions make it tractable.

3.1 Reduction 1: superposition for the space charge

Write

$$\phi(x,y) = \phi_p(y) + \phi_h(x,y), \tag{5}$$

where ϕ_p is the exact 1-D vertical solution carrying *all* the charge,

$$\frac{\mathrm{d}}{\mathrm{d}y}\Big(\varepsilon(y)\,\frac{\mathrm{d}\phi_p}{\mathrm{d}y}\Big) = -\rho(y), \qquad \phi_p(0)=V, \quad \phi_p(H)=0. \tag{6}$$

Integrating once, with the accumulated charge $Q(y) = \int_0^y \rho\,\mathrm{d}y'$,

$$\phi_p'(y) = \frac{D_0 - Q(y)}{\varepsilon(y)}, \qquad D_0 = \frac{\displaystyle\int_0^H \frac{Q(y)}{\varepsilon(y)}\mathrm{d}y - V}{\displaystyle\int_0^H \frac{\mathrm{d}y}{\varepsilon(y)}}, \tag{7}$$

D_0 being fixed by $\phi_p(H) = 0$. Substituting (5) into (1) and using (6) leaves ϕ_h *source-free*: $\nabla\cdot(\varepsilon\nabla\phi_h) = 0$. Because ϕ_p carries the donor charge, it is what pushes s_{ox} past $\frac{1}{2}$.

3.2 Reduction 2: effective oxide thickness

Define the effective depth

$$Y(y) = \begin{cases} y, & y \le t_{\mathrm{si}}, \\ t_{\mathrm{si}} + \dfrac{\varepsilon_{\mathrm{si}}}{\varepsilon_{\mathrm{ox}}}\,(y - t_{\mathrm{si}}), & y > t_{\mathrm{si}}, \end{cases} \qquad H_{\mathrm{eff}} = t_{\mathrm{si}} + \frac{\varepsilon_{\mathrm{si}}}{\varepsilon_{\mathrm{ox}}} t_{\mathrm{ox}} = 29\,\mu\mathrm{m}. \tag{8}$$

This stretch makes the normal displacement continuous automatically, collapsing the two-dielectric source-free equation into a single-dielectric Laplace equation $\partial_{xx}\phi_h + \partial_{YY}\phi_h = 0$ on the rectangle $[0, L_x] \times [0, H_{\mathrm{eff}}]$. It is exact for vertical fields (which dominate the thin oxide) and is the method's sole approximation.

3.3 The Schwarz–Christoffel map

Map that rectangle to the upper half w-plane with the Jacobi elliptic sine,

$$w = \mathrm{sn}(u, m), \qquad u = \frac{2K(m)}{L_x}\,(x + \mathrm{i}Y) - K(m), \tag{9}$$

whose modulus m is fixed by matching the rectangle aspect ratio,

$$\frac{K'(m)}{K(m)} = \frac{2H_{\mathrm{eff}}}{L_x} = 1.16 \quad \Longrightarrow \quad m = 0.342058, \tag{10}$$

K, K' being the complete elliptic integrals. The complex argument is evaluated through the imaginary-argument addition theorem (Abramowitz & Stegun 16.21); with $s, c, d = \mathrm{sn}, \mathrm{cn}, \mathrm{dn}(\alpha, m)$ and $s_1, c_1, d_1 = \mathrm{sn}, \mathrm{cn}, \mathrm{dn}(\beta, 1-m)$,

$$\mathrm{sn}(\alpha + \mathrm{i}\beta, m) = \frac{s\,d_1 + \mathrm{i}\,c\,d\,s_1 c_1}{c_1^2 + m\,s^2 s_1^2}. \tag{11}$$

3.4 Boundary data and harmonic measure

The four rectangle sides map to the real axis; the grounded substrate maps to $|t| > 1/\sqrt{m}$ and (being $\phi_h = 0$) contributes nothing. The Dirichlet data for ϕ_h are $\phi_{\mathrm{actual}} - V$ on the top (drain $\to 0$, source $\to -V$, cosine ramp between), $-\phi_p(y)$ on the p^+ edge, and 0 on the left edge. At an interior image point w the solution is the Poisson integral, i.e. the harmonic measure of the boundary data,

$$\phi_h(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} g(t)\,\frac{\mathrm{Im}\,w}{(\mathrm{Re}\,w - t)^2 + (\mathrm{Im}\,w)^2}\,\mathrm{d}t = \frac{1}{\pi}\sum_i \bar{g}_i\,\Delta\theta_i, \tag{12}$$

where $\Delta\theta_i$ is the angle subtended at w by boundary segment i and $\bar{g}_i$ its mean value. No PDE grid is used. The full field is $\phi = \phi_p(y) + \phi_h(x, Y(y))$.

3.5 Algorithm

1. Solve the 1-D profile $\phi_p(y)$ in closed form, (7).
2. Find the modulus m from (10) by a bracketed root find on $K'(m)/K(m)$.
3. Sample the three active rectangle sides; map each sample to the real axis via (9)–(11) and tag it with its Dirichlet value g.
4. For every output point, evaluate ϕ_h by the subtended-angle sum (12); add ϕ_p.

This is `fig4_conformal.py`. *Verification:* ϕ_h obeys the mean-value property to 8×10^{-16} (machine precision), boundary conditions are recovered to $< 0.1\,\mathrm{V}$, and $s_{\mathrm{ox}} = 0.582$.

4 Method 3 — Green's function (eigenfunction series)

The third method uses the Dirichlet *Green's function* of the rectangle, handling the space charge *directly* by a volume integral (no ϕ_p trick). Working again in effective coordinates (x, Y) on $[0, a] \times [0, b]$, $a = L_x$, $b = H_{\mathrm{eff}}$, Poisson's equation is $\nabla^2\phi = -f$ with $f = \rho(Y)/\varepsilon_{\mathrm{si}}$ in the silicon band $Y < t_{\mathrm{si}}$ and 0 elsewhere.

4.1 The rectangle Green's function

The Green's function satisfying $\nabla^2 G = -\delta$ with $G = 0$ on all four walls is the eigenfunction series

$$G(x, Y; x', Y') = \frac{4}{ab} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{\sin \frac{n\pi x}{a} \sin \frac{n\pi x'}{a} \sin \frac{m\pi Y}{b} \sin \frac{m\pi Y'}{b}}{\left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2}. \tag{13}$$

The complete solution is then Green's third identity,

$$\phi(\mathbf{r}) = \underbrace{\int_V G\, f \, \mathrm{d}V'}_{u} \; - \; \underbrace{\oint_S \phi_{\mathrm{bc}} \frac{\partial G}{\partial n} \, \mathrm{d}S'}_{B+v'}, \tag{14}$$

a volume term for the charge plus a boundary term for the electrode data.

4.2 The three series

(u) space-charge response. Because f depends on Y only, its x-expansion is that of a constant (odd modes), giving

$$u(x, Y) = \sum_{n,m} u_{nm} \sin \tfrac{n\pi x}{a} \sin \tfrac{m\pi Y}{b}, \quad u_{nm} = \frac{b_n e_m}{(n\pi/a)^2 + (m\pi/b)^2}, \tag{15}$$

with $b_n = \dfrac{4}{n\pi}$ (n odd, else 0) and $e_m = \dfrac{2}{b} \displaystyle\int_0^{t_{\mathrm{si}}} \frac{\rho(Y)}{\varepsilon_{\mathrm{si}}} \sin \tfrac{m\pi Y}{b} \, \mathrm{d}Y$. u vanishes on all walls and is the potential of the trapped donor charge in a grounded box.

(B) corner-carrying bilinear. The drain touches the top-left corner at V; the harmonic bilinear

$$B(x, Y) = V\Big(1 - \tfrac{x}{a}\Big)\Big(1 - \tfrac{Y}{b}\Big), \qquad \nabla^2 B = 0, \tag{16}$$

carries that single non-zero corner so the remaining wall data vanish at every corner and their sine series converge cleanly.

(v′) reduced electrode data. Only the top and left walls carry non-zero reduced data:

$$\phi_h^{\mathrm{top}} = \sum_n A_n \sin \tfrac{n\pi x}{a} \frac{\sinh \frac{n\pi(b-Y)}{a}}{\sinh \frac{n\pi b}{a}}, \qquad A_n = \frac{2}{a} \int_0^a \big[\phi_{\mathrm{top}}(x) - V(1 - \tfrac{x}{a})\big] \sin \tfrac{n\pi x}{a} \, \mathrm{d}x, \tag{17}$$

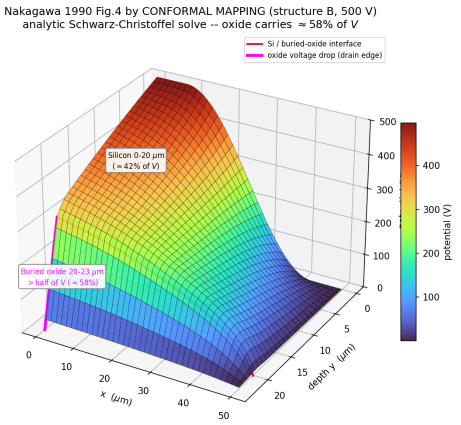
$$\phi_h^{\mathrm{left}} = \sum_m C_m \sin \tfrac{m\pi Y}{b} \frac{\sinh \frac{m\pi(a-x)}{b}}{\sinh \frac{m\pi a}{b}}, \qquad C_m = \frac{2}{b} \int_0^b \big[\phi_p(y(Y)) - V(1 - \tfrac{Y}{b})\big] \sin \tfrac{m\pi Y}{b} \, \mathrm{d}Y. \tag{18}$$

The sinh ratios are evaluated in exponential form to avoid overflow at large mode number. The full field is $\phi = u + B + \phi_h^{\mathrm{top}} + \phi_h^{\mathrm{left}}$; one checks it satisfies every wall condition term-by-term.

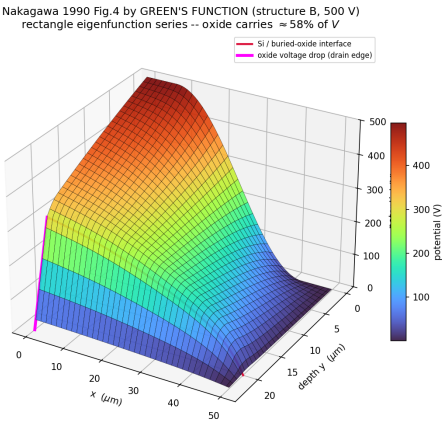
4.3 Algorithm

1. Precompute the charge coefficients e_m and the wall coefficients A_n, C_m by numerical quadrature of (15)–(18).
2. Form the basis matrices $\sin(n\pi x/a)$, $\sin(m\pi Y/b)$ on the output grid.
3. Sum the three series (as dense matrix products) and add B; truncate at $N = 400$ modes.

This is `fig4_green.py`. *Verification:* boundary conditions recovered to 0.00 V, effective-coordinate Poisson residual RMS $\sim 0.5\,(V/H_{\mathrm{eff}}^2)$, and $s_{\mathrm{ox}} = 0.582$ — identical to the conformal map.



(a) Conformal map — 3-D potential.



(b) Green's function — 3-D potential.

Figure 2: The Fig. 4 potential surface from the two analytic methods. Both show the gentle slope through the silicon and the steep "voltage cliff" across the buried oxide at the drain edge (magenta), carrying $\approx 58\%$ of V.

Table 2: Oxide voltage share s_{ox} under the drain, and cross-checks.

Method	s_{ox}	note
Finite-volume Poisson (Method 1)	0.539	grid-converged ≈ 0.54; the yardstick
Conformal map (Method 2)	0.582	ϕ_h harmonic to 8×10^{-16}
Green's function (Method 3)	0.582	BCs exact; residual $\sim 0.5\,V/H_{\mathrm{eff}}^2$
Charge-free divider	0.29–0.31	Laplace limit; dielectric ratio only
Paper (Fig. 4)	> 0.5	"more than half"

5 Results and plots

5.1 Discussion

Three observations close the loop:

- **The two analytic methods agree to** 0.09 V **RMS.** The conformal map (elliptic functions, harmonic measure) and the Green's function (eigenfunction series) solve the *identical* idealised boundary-value problem by unrelated mathematics; their coincidence (Fig. 3) validates both independently of the finite-volume solve.
- **The residual** $\sim 5\%$ **gap to finite volumes is the shared boundary idealisation**, not a defect of either analytic method: it lives in near-surface dipoles at the electrode edges (Fig. 4b), exactly where the analytic Dirichlet closures depart from the physical Neumann free surface.
- **The space charge is the physics.** The charge-free divider gives only $t_{\mathrm{ox}}/\varepsilon_{\mathrm{ox}}/(t_{\mathrm{si}}/\varepsilon_{\mathrm{si}} + t_{\mathrm{ox}}/\varepsilon_{\mathrm{ox}}) \approx 0.31$; the donor sheet lifts this by nearly 30 points to $> \frac{1}{2}$ (Fig. 4a), reproducing the paper's headline result.

6 Reproducing the figures

Each script is a standalone Python 3 entry point (numpy, scipy, matplotlib):

```
cd nakagawa1990_repro/code
python fig4_pde_standalone.py   # Method 1: finite-volume Poisson  -> fig4_pde_standalone_3d.png
python fig4_conformal.py        # Method 2: conformal mapping      -> fig4_conformal*.png
python fig4_green.py            # Method 3: Green's function + 3-method compare
                                #           -> fig4_green*.png, fig4_three_methods.png
```

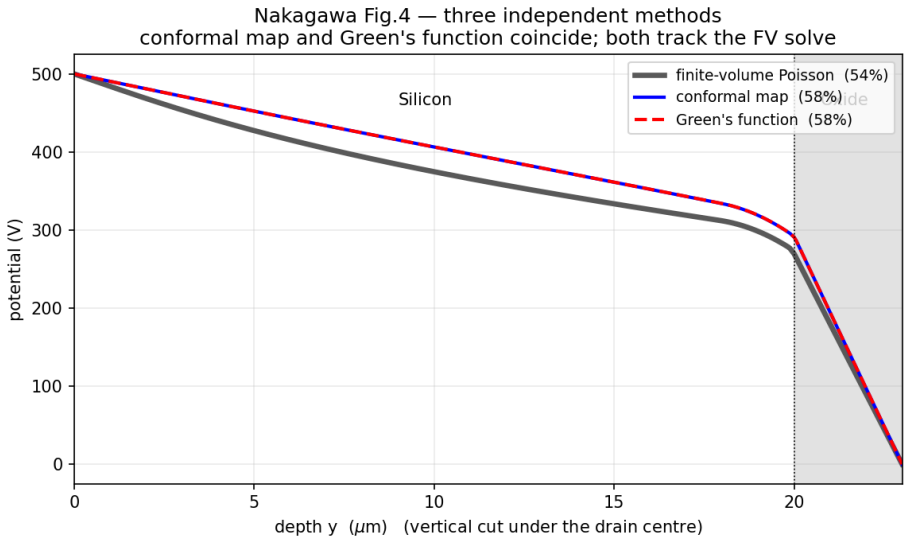
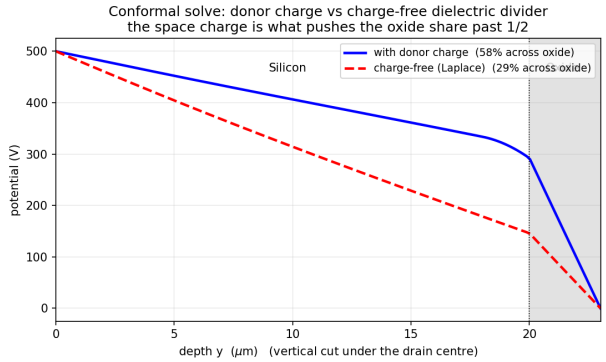
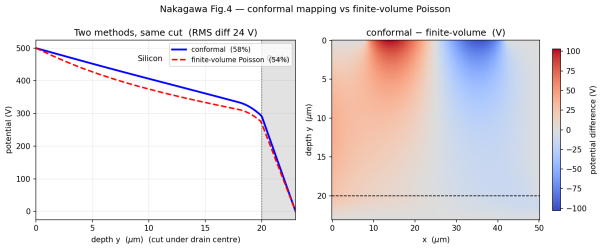


Figure 3: Vertical cut under the drain centre for all three 2-D methods. The conformal-map (blue) and Green's-function (red dashed) curves are visually indistinguishable (RMS difference 0.09 V); both track the finite-volume Poisson solve (grey), differing only by the shared boundary-condition idealisation in the silicon and converging across the oxide.



(a) Donor charge vs charge-free.



(b) Conformal − finite-volume difference.

Figure 4: (a) Removing the space charge collapses the oxide share from 58% to 29%: the donor sheet, not the dielectric ratio alone, produces the "more than half" result. (b) The analytic−numeric difference is confined to near-surface dipoles at the electrode edges (the BC idealisation); the bulk and the entire oxide agree.

Numbers are printed next to the paper's values by each script's `verify()`; figures are written to `figs/`. Compile this report with `pdflatex fig4_multimethod_report.tex` from `report/`.